

Experiment – 1 B

```
[1]
✓ 6s import nltk
      nltk.download('brown')
      nltk.download('universal_tagset')
```

▼

```
[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data]   Unzipping corpora/brown.zip.
[nltk_data] Downloading package universal_tagset to /root/nltk_data...
[nltk_data]   Unzipping taggers/universal_tagset.zip.
True
```

```
[4]
✓ 1s ▶ from nltk.corpus import brown
      sentences=brown.sents()
      tokens=list(word.lower() for sent in sentences for word in sent)
      print("Total sentences:",len(sentences))
      print("Total tokens:",len(tokens))
      tokens[:20]
```

▼

```
... Total sentences: 57340
Total tokens: 1161192
['the',
 'fulton',
 'county',
 'grand',
 'jury',
 'said',
 'friday',
 'an',
 'investigation',
 'of',
 'atlanta's',
 'recent',
 'primary',
 'election',
 'produced',
 '\'',
 'no',
 'evidence',
 '\"\",
 'that']
```

[5]
✓ 5s

```
from nltk.tag import hmm
from collections import defaultdict, Counter
train_data=brown.tagged_sents(tagset='universal')
trainer=hmm.HiddenMarkovModelTrainer()
hmm_model=trainer.train_supervised(train_data)
print("HMM Model Trained Successfully")
```

▼ HMM Model Trained Successfully

[7]
✓ 7s

```
import math
pos_sequences=[[tag for (_,tag) in sent] for sent in train_data]
pos_bigram=defaultdict(Counter)
for seq in pos_sequences:
    for i in range(len(seq)-1):
        pos_bigram[seq[i]][seq[i+1]]+=1
word_by_tag=defaultdict(Counter)
for sent in train_data:
    for word, tag in sent:
        word_by_tag[tag][word.lower()]+=1
def hmm_predict_next(sentence):
    words=sentence.lower().split()
    tagged=hmm_model.tag(words)
    last_tag=tagged[-1][1]
    next_pos=pos_bigram[last_tag].most_common(1)[0][0]
    candidates=word_by_tag[next_pos].most_common(5)
    return next_pos,candidates
pos,words=hmm_predict_next("the government decided")
print("Predicted POS:",pos)
print("Suggested Next Words:",words)
```

... Predicted POS: VERB
Suggested Next Words: [('is', 10108), ('was', 9815), ('be', 6374), ('had', 5132), ('are', 4394)]